2024

COMPUTER SCIENCE

Paper : CSMC-104

(Object Oriented Analysis and Design)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** from the following : 2×5

   (a) What is the purpose of the final keyword in Java?

   (b) Explain the use of static variables.

   (c) What is the distinction between an Error and an Exception in Java?

   (d) Name two Structural UML diagrams and two Behavioural UML diagrams.

   (e) How is the generalization relationship represented in a class diagram?

   (f) What is meant by a lost message and how is it represented in a sequence diagram?

   (g) What are the advantages of using design patterns?

2. Answer ***any five*** from the following :

   (a) What is multiple inheritance? Explain the differences between method hiding and method overriding in Java. 1+3

   (b) Why high cohesion and loose coupling are considered desirable characteristics for a system? 4

   (c) Explain how Java facilitates dynamic method dispatch. 4

   (d) How does a class diagram represent different relationships between classes? Provide examples for each relationship. 4

   (e) How does synchronous and asynchronous message transfer occur in a Sequence diagram? Explain with examples. 2+2

   (f) What are the different components of the Adapter design pattern? Explain the role of each component. 1+3

   (g) Name the different types of design patterns and provide one example for each type. What are the advantages of using the Chain of Responsibility design pattern? 2+2

3. Answer ***any four*** of the following :

   (a) Define a Java interface. How does it differ from an abstract class? Provide a scenario where an interface is better than an abstract class. How is runtime polymorphism achieved in Java? Explain with an example. 2+4+4

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(b) What are the different types of constructors available in Java? Explain each type with an example. How can a class be prevented from being inherited in Java? Demonstrate this concept using an example. 1+3+6

(c) Consider an RFID-based warehouse system where RFID tags are attached to goods, pallets and storage bins and RFID readers are placed throughout the warehouse. The system captures and processes data automatically as tagged items move through various stages of the warehouse lifecycle, from receiving to storage, retrieval and shipping. Real-time data on stock levels, inventory movement and asset tracking is crucial for making informed business decisions, improving operational efficiency and reducing losses.

Create a class diagram that captures the entities, their relationships and the key functionalities of the RFID-based warehouse management system. 10

(d) What are the different relationships between the actors and use cases in a Use Case diagram? Explain each of them. Consider an Online Shopping System containing Customers and Admin. Customers can check-in, browse products, add products to the cart and check out. The Admin is enabled to manage and change the inventory. Draw a Use Case diagram to show the system's different operations. If you are considering any specific assumptions, mention them. 4+6

(e) Explain how the Factory Method design pattern can be used as a virtual constructor. How is the Abstract Factory Method different from the Factory Method design pattern? Describe how early and lazy installation occurs in the Singleton design pattern. 4+2+4

(f) Answer *any four* of the following : 2½×4

(i) Function overloading in Java

(ii) Use of Guard Conditions in Sequence diagrams

(iii) Multiplicity of association in Class diagrams

(iv) Fork and Join operations in Activity diagrams

(v) Sequence fragments in Sequence diagrams

(vi) When to use the Flyweight Design Pattern.